Measurements Final Project

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December 11, 2025

https://www.youtube.com/watch?v=sOnj7LjEwes

Introduction

- Eddy Current Tachometer
  - Device used to measure RPM of a shaft or rotating system
  - Uses electromagnetic induction → no mechanical contact needed
  - Common in engines and machinery
- System to be divided into:
  - *Actuation Module* – controls shaft rotation
  - *Measurement Module* – uses eddy current principle for RPM sensing
  - *Display Module* – mechanical pointer + digital readout
- Device calibrated using a reference tachometer

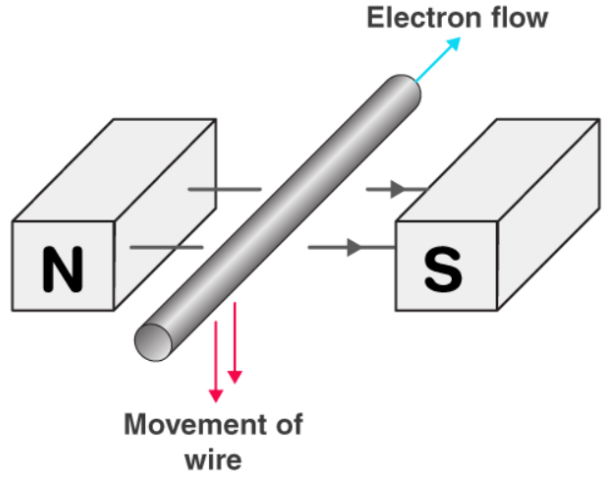


Theory

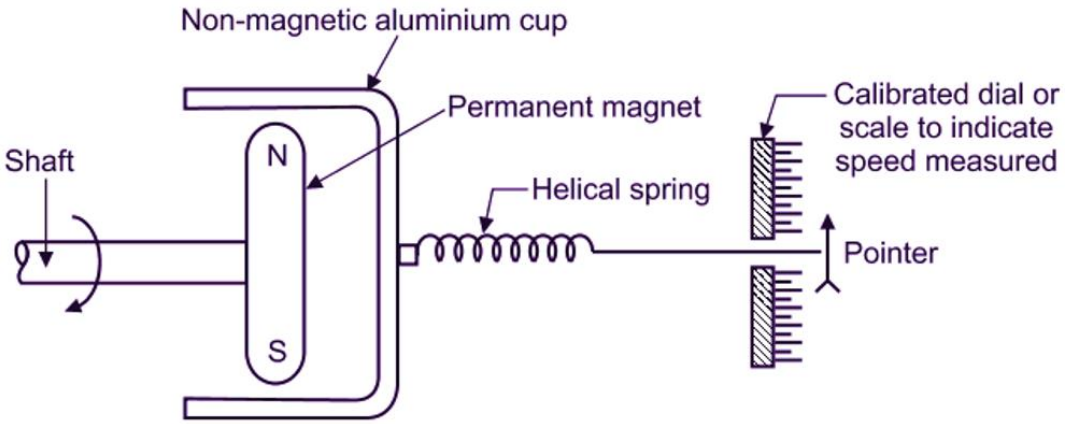
- Electromagnetic Induction
  - Rotating conductive disk moves through a magnetic field
  - Changing magnetic flux induces eddy currents in disk
  - Based on Faraday's Law of Induction
- Lenz's Law and Opposing Torque
  - Eddy currents create their own magnetic field
    - Field opposes the motion that creates it
  - Results in a drag torque that increases with speed
- Tachometer
  - Drag torque transmitted to spindle connected to a return spring
  - Higher RPM = stronger induced torque = larger pointer deflection
  - Pointer angle roughly proportional to rotational speed

$$\theta \propto \omega$$

- Limitations
  - Lower sensitivity at very lower speeds
  - Must be calibrated for accurate RPM
    - Compare device output to known RPM (laser tachometer)
    - Will ensure mechanical dial & digital display show corrected values



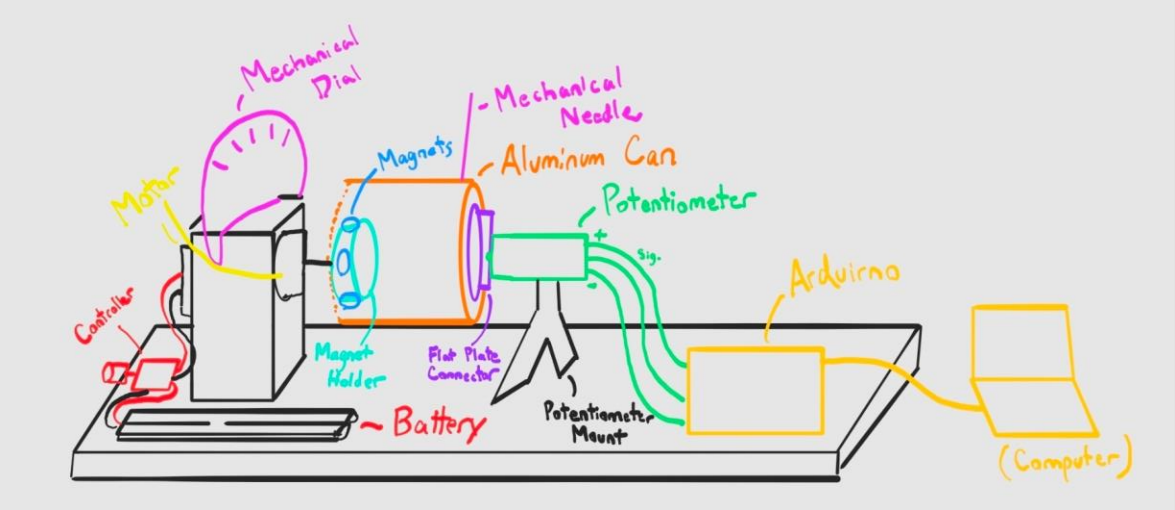
Visual Representation of Lenz's Law



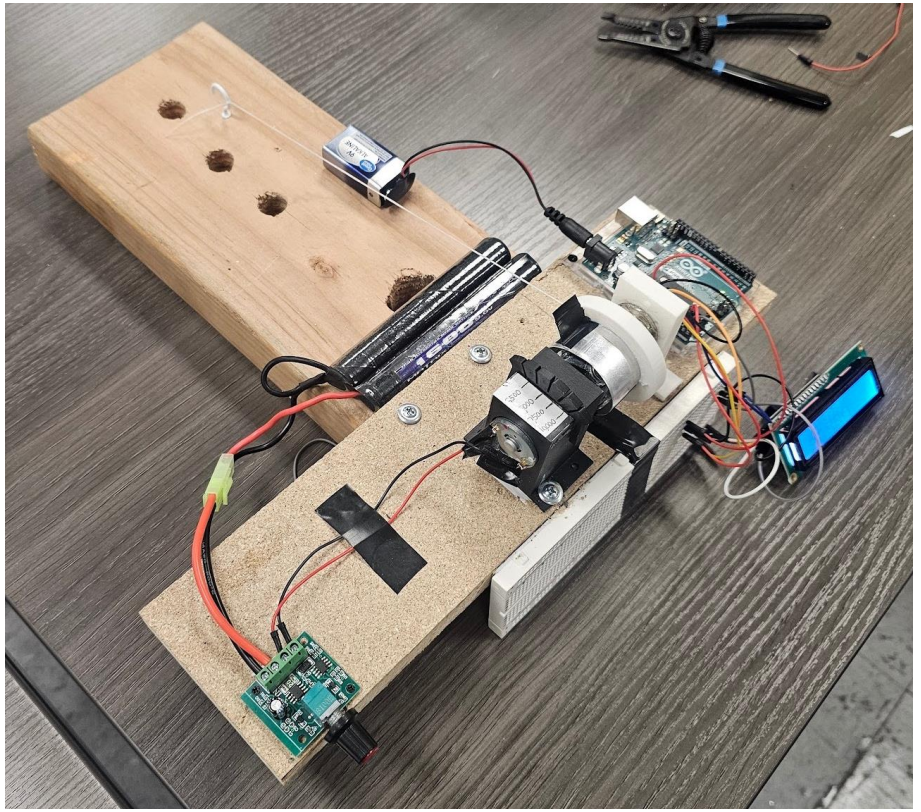
Sample Eddy Current Tachometer

Design / Experiment

- Magnet and aluminum drag clearance
  - 3mm-5mm



Initial Concept of Eddy Current Tachometer

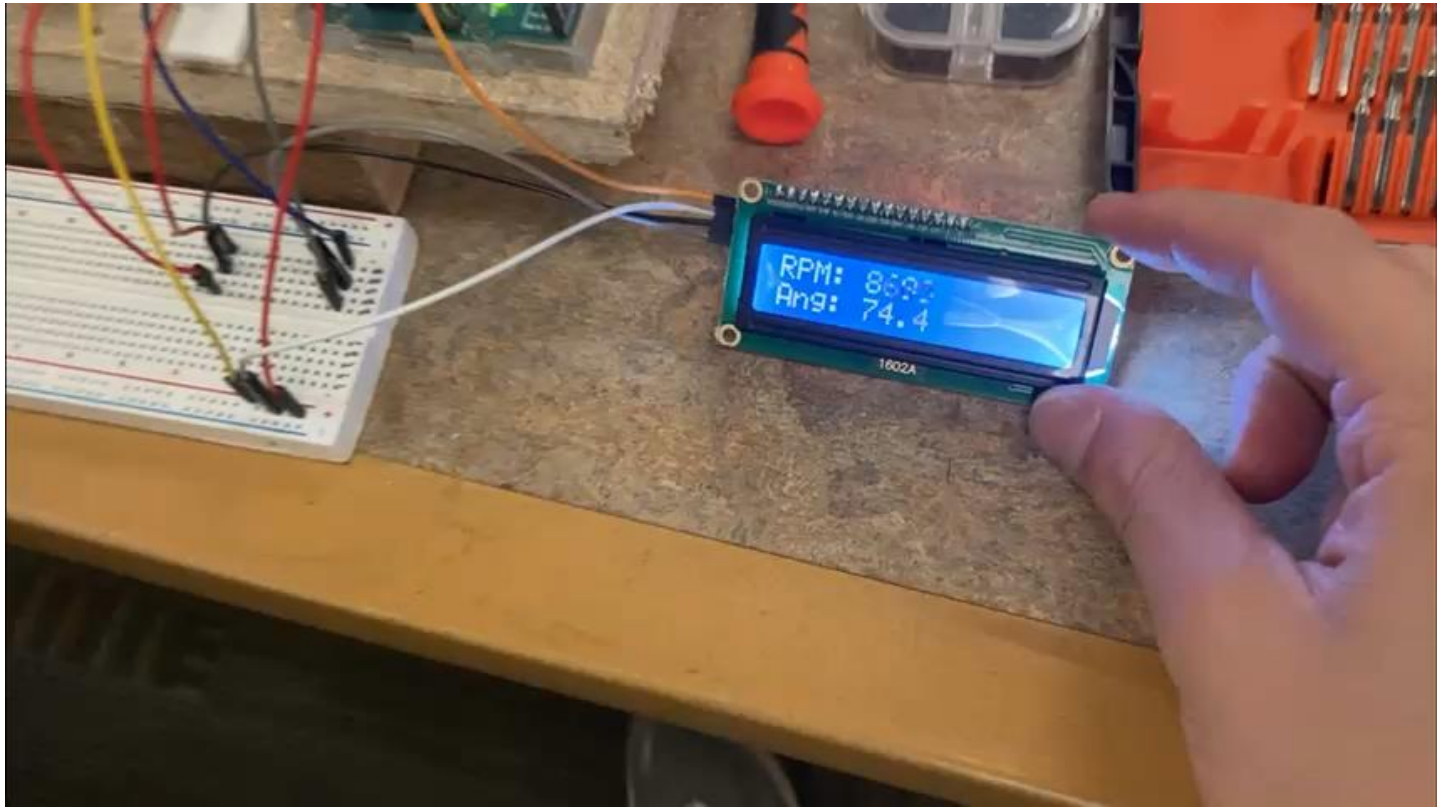


Final Concept of Eddy Current Tachometer

Bill of Materials

PWM motor control knob:	*$2.85*
9.6V Electric motor:	*$8.99*
Arduino uno:	*$26.79*
9.6V Nunchuck battery:	*$16.00*
AS5600 magnetic encoder:	*$2.50*
Filament:	*$3.00*
Aluminum Flash:	*$0.01*
Ball Bearing:	*$0.60*
Hardware:	*$1.75*
Total:	**$62.49**

Demonstration Video



Working Video

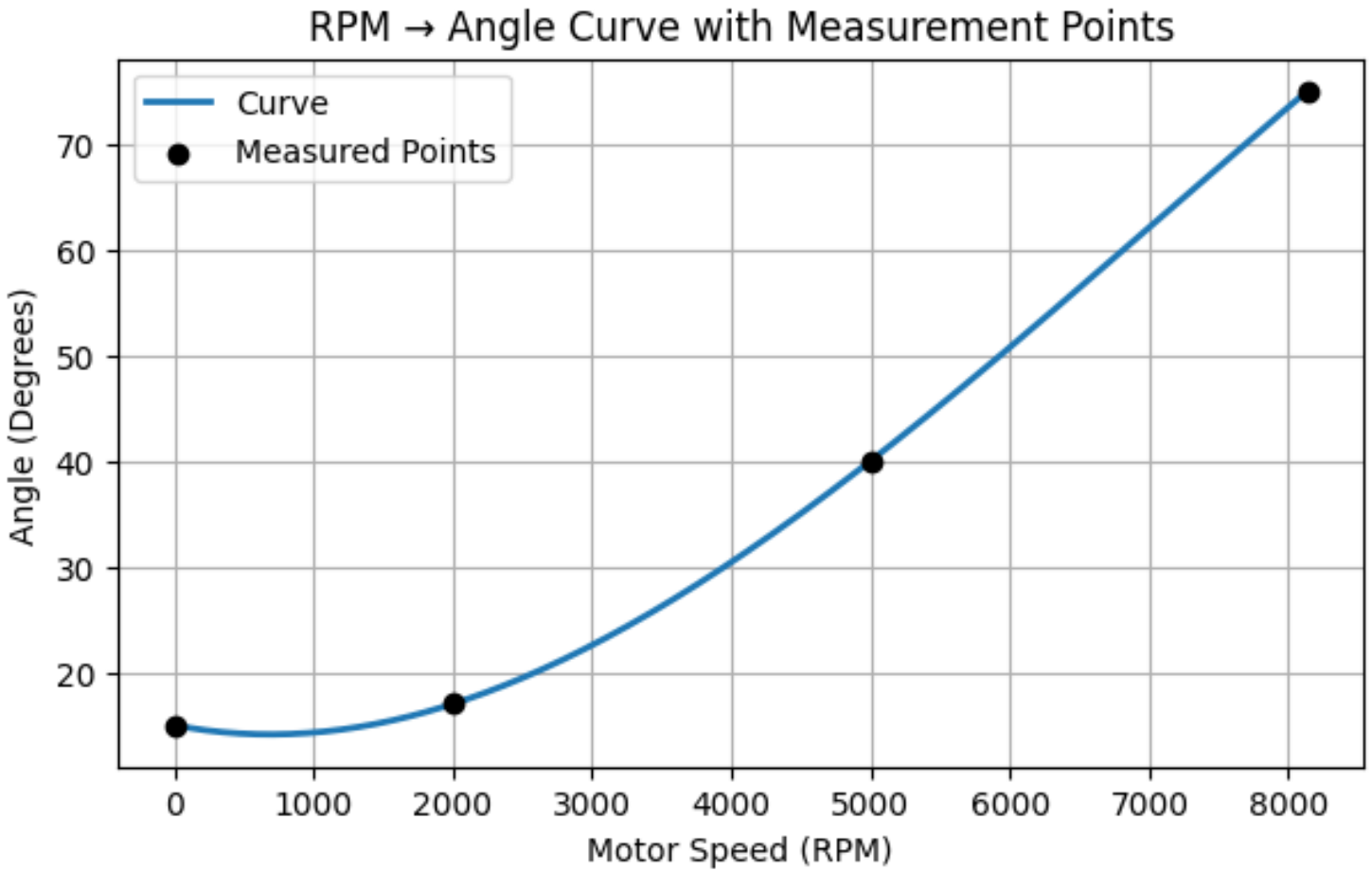
Evaluation



Calibration Recording

Calibration

- (0, 15.2)
- (2000, 17.8)
- (5000, 40.5)
- (8136, 75.3)



Conclusion

- We built a working eddy-current tachometer that can measure the motor's RPM and show it on both a digital LCD and an analog drag-cup indicator.
- Along the way, we learned how important the spacing is between the rotating magnet and the aluminum drag cup, too close or too far and the readings become inconsistent.
- Calibration turned out to be one of the most important steps. Getting a good match between RPM and the pointer's angle was necessary before the readings made sense.
- The current setup works as a prototype, but there's room to improve it. Some ideas include adding more calibration points to tighten accuracy, adjusting the drag-cup design, and improving the signal-processing circuit so the readings respond more smoothly.

References

Mohan, N., Robbins, W. P., & Undeland, T. M. (2003). *Power Electronics: Converters, Applications, and Design*. Wiley. (For eddy current electromagnetic behavior)

Greenwood, D. T. (1997). *Principles of Dynamics*. Prentice Hall. (For torque and rotational dynamics relationships)